

Z-Test vs. T-Test: Concept, Differences, and Numerical Examples

1. Introduction

Both Z-test and T-test are statistical hypothesis tests used to compare sample data with population data or between two groups. The choice between them depends on **sample size** and whether the **population standard deviation (σ)** is known.

2. Key Differences Between Z-Test and T-Test

Criteria	Z-Test	T-Test
Definition	Used to compare sample and population means when population standard deviation (σ) is known.	Used when the sample size is small ($n < 30$) and population standard deviation (σ) is unknown.
Sample Size	Large samples ($n \geq 30$)	Small samples ($n < 30$)
Standard Deviation	Known (σ is given)	Unknown (σ is estimated using the sample standard deviation s)
Distribution	Follows normal distribution	Follows t-distribution
Formula	$Z = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}}$	$t = \frac{\bar{X} - \mu}{s / \sqrt{n}}$
Used In	Large-scale studies, known population parameters	Small studies, unknown population parameters
Example	Checking if a factory's machine produces bottles with an average of 500ml .	Checking if a new diet program reduces weight in a small group of 10 people.

3. Z-Test: Concept and Example

3.1 Concept

A Z-test is used when:

- The sample size is **large** ($n \geq 30$).

2. The **population standard deviation** (σ) is known.
3. The data follows a **normal distribution**.

3.2 Formula

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

where:

- $\bar{X}$ = Sample mean
- μ = Population mean
- σ = Population standard deviation
- n = Sample size

3.3 Example: Z-Test for Population Mean

Problem:

A machine is supposed to fill bottles with **500ml** of water. A quality control team takes a **random sample of 40 bottles** and finds an **average of 495ml** with a **known population standard deviation of 10ml**. Test at $\alpha = 0.05$ if the machine is filling correctly.

Step 1: State Hypotheses

- $H_0 : \mu = 500$ (No difference in mean bottle filling)
- $H_a : \mu \neq 500$ (The machine is underfilling or overfilling)

Step 2: Compute Z-Score

$$Z = \frac{495 - 500}{10/\sqrt{40}}$$

$$Z = \frac{-5}{10/6.32} = \frac{-5}{1.58} = -3.16$$

Step 3: Compare with Critical Value

For $\alpha = 0.05$ (two-tailed test), the **critical Z-value** = ± 1.96 .

Since $Z = -3.16 < -1.96$, we **reject** H_0 .

Step 4: Conclusion

The machine is **not filling correctly** as the mean differs significantly from 500ml.

4. T-Test: Concept and Example

4.1 Concept

A **T-test** is used when:

1. The sample size is **small** ($n < 30$).
2. The **population standard deviation** (σ) is **unknown**.
3. The data follows a **t-distribution**.

4.2 Formula

$$t = \frac{\bar{X} - \mu}{s/\sqrt{n}}$$

where:

- s = Sample standard deviation (since σ is unknown)
- Degrees of freedom (df) = $n - 1$

4.3 Types of T-Tests

Type	Purpose	Example
One-Sample T-Test	Compares sample mean to population mean	Checking if a class's average test score differs from 70
Independent T-Test	Compares means of two independent groups	Comparing males' vs. females' average salaries
Paired T-Test	Compares the same group before and after treatment	Checking if a weight loss program is effective

4.4 Example: One-Sample T-Test

Problem:

A fitness trainer claims that a new **diet reduces weight by 5kg** in a month. A test on **10 participants** shows an **average weight loss of 4.2kg** with a **sample standard deviation of 1.5kg**. Test at $\alpha = 0.05$ if the diet is effective.

Step 1: State Hypotheses

- $H_0 : \mu = 5$ (Diet causes 5kg weight loss)
- $H_a : \mu < 5$ (Diet is **less effective**)

Step 2: Compute T-Statistic

$$t = \frac{4.2 - 5}{1.5/\sqrt{10}}$$

$$t = \frac{-0.8}{1.5/3.16} = \frac{-0.8}{0.474} = -1.69$$

Step 3: Find Critical Value

- For $\alpha = 0.05$ (one-tailed) and $df = 9$, $t_{critical} = -1.833$.

Step 4: Compare

Since $t = -1.69$ is greater than $t_{critical} = -1.833$, we fail to reject H_0 .

Step 5: Conclusion

The diet does not significantly reduce weight below 5kg.

5. Summary Table

Feature	Z-Test	T-Test
When to Use	Large samples ($n \geq 30$), known σ	Small samples ($n < 30$), unknown σ
Formula	$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$	$t = \frac{\bar{X} - \mu}{s/\sqrt{n}}$
Critical Value Table	Z-table	T-table (df = $n - 1$)
Example	Checking if a factory's machine fills bottles correctly	Checking if a new drug reduces blood pressure

6. When to Use Z-Test vs. T-Test

- Use a Z-Test when:
 - Sample size $n \geq 30$.
 - Population standard deviation σ is known.
 - Data follows a normal distribution.
- Use a T-Test when:
 - Sample size $n < 30$.
 - Population standard deviation σ is unknown.
 - Data is approximately normal.

7. Final Thoughts

- Z-test is for large samples, t-test is for small samples.
- T-tests are more flexible since they work with **unknown standard deviations**.
- Both tests help in **decision-making** by comparing a sample to a population.